

Akash Kondamoeina

Phone Number | Email | LinkedIn | Portfolio | Github

Professional Summary

AI/ML Engineer with 4+ years of experience designing, developing, and deploying scalable Artificial Intelligence, Machine Learning, and Generative AI solutions for enterprise applications. Proficient in Python, SQL, TensorFlow, PyTorch, Scikit-learn, Pandas, and NumPy, with expertise in building predictive models, deep learning solutions, and end-to-end machine learning pipelines, including data preprocessing, feature engineering, model training, evaluation, deployment, and monitoring. Experienced in developing NLP and Generative AI applications using Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, OpenAI, and Azure OpenAI to deliver intelligent automation and conversational AI solutions. Skilled in deploying production-ready AI applications using Docker, Kubernetes, FastAPI, MLflow, Apache Airflow, and CI/CD pipelines across AWS, Microsoft Azure, and Google Cloud Platform (GCP). Hands-on experience with Amazon SageMaker, Azure Machine Learning, Google Vertex AI, vector databases such as Pinecone and FAISS, and MLOps best practices to build secure, scalable, and high-performance AI solutions. Strong analytical, problem-solving, and collaboration skills with a passion for leveraging emerging AI technologies to deliver innovative, reliable, and business-driven intelligent systems.

Core Skills

Programming Languages:	Python, SQL, PySpark
Machine Learning:	Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, Feature Engineering, Model Training, Model Evaluation, Hyperparameter Tuning
Generative AI & LLMs:	OpenAI GPT, LLaMA, Claude, Gemini, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Integration
Natural Language Processing (NLP):	Transformers, BERT, Hugging Face Transformers, Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Semantic Search, Text Summarization.
AI Frameworks & Vector Databases:	LangChain, Hugging Face, Pinecone, FAISS, ChromaDB
Cloud & MLOps:	AWS (S3, EC2, SageMaker, Glue, Lambda, Redshift), Azure Machine Learning, Google Vertex AI, Docker, Kubernetes, MLflow, Apache Airflow, GitHub Actions, CI/CD
Programming & Data Engineering:	FastAPI, REST APIs, Pandas, NumPy, Kafka, dbt
Responsible AI & Model Optimization:	Model Monitoring, Explainable AI (XAI), Bias Detection, Model Performance Optimization

Professional Experience

AI/ML Engineer | Meijer | Chicago, IL, USA

Sep 2024 – Present

- Designed and deployed scalable AI/ML solutions on AWS to support demand forecasting, customer behavior analytics, inventory optimization, personalized recommendations, and intelligent retail operations across enterprise applications.
- Developed scalable data pipelines using Amazon S3, AWS Glue, AWS Lambda, and Amazon Redshift to ingest, process, and transform large-scale customer, product, sales, and inventory datasets for machine learning applications.
- Built Generative AI applications using Large Language Models (LLMs), Prompt Engineering, and Retrieval-Augmented Generation (RAG) with LangChain, OpenAI, and Hugging Face Transformers to deliver AI-powered customer support, knowledge retrieval, and intelligent product search solutions.

- Developed Natural Language Processing (NLP) models for sentiment analysis, text classification, document summarization, semantic search, and customer feedback analysis to enhance customer engagement and business decision-making.
- Designed and optimized recommendation engines and personalization models using machine learning algorithms to improve customer experience, product discovery, and targeted marketing campaigns.
- Built and deployed production-ready AI models using Amazon SageMaker, Docker, Kubernetes, and FastAPI, enabling scalable, secure, and high-performance real-time inference across cloud environments.
- Integrated vector databases including Pinecone and FAISS with embedding models to develop intelligent enterprise search and AI-powered knowledge management solutions for retail operations.
- Implemented MLOps best practices using MLflow, Apache Airflow, GitHub Actions, and CI/CD pipelines to automate model training, deployment, monitoring, and continuous performance optimization.
- Monitored model performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and model drift metrics, applying hyperparameter tuning and Explainable AI (XAI) techniques to improve reliability, transparency, and business impact.
- Collaborated with product managers, data engineers, business analysts, and cross-functional teams to design, deploy, and enhance AI-driven retail solutions that improved operational efficiency, customer satisfaction, and data-driven decision-making.

Environment: Python, SQL, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Hugging Face Transformers, LangChain, OpenAI, Azure OpenAI, FastAPI, Pinecone, FAISS, AWS (S3, EC2, SageMaker, Glue, Lambda, Redshift), Docker, Kubernetes, MLflow, Apache Airflow, GitHub Actions, Tableau, Power BI.

Data scientist | Enter-Pi Software Solutions | Hyderabad

Sep 2020 – Dec 2023

- Designed and developed end-to-end machine learning solutions for enterprise clients using Python, SQL, Scikit-learn, TensorFlow, and PyTorch to solve business problems through predictive analytics and intelligent automation.
- Built scalable data pipelines and ETL workflows using AWS Glue, AWS Lambda, Python, and SQL to ingest, cleanse, transform, and process structured and unstructured data from multiple enterprise applications.
- Performed exploratory data analysis (EDA), feature engineering, and statistical modeling to prepare high-quality datasets and improve the performance and accuracy of machine learning models.
- Developed supervised and unsupervised machine learning models, including classification, regression, clustering, and anomaly detection algorithms, to support business forecasting, customer analytics, and operational optimization.
- Designed and implemented Natural Language Processing (NLP) solutions using Hugging Face Transformers for text classification, sentiment analysis, document intelligence, and information extraction from unstructured data.
- Built RESTful APIs using FastAPI and deployed production-ready machine learning models using Docker, Kubernetes, AWS SageMaker, and Apache Airflow to enable scalable AI solutions across enterprise environments.
- Evaluated, optimized, and monitored machine learning models using performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and RMSE, ensuring reliable and high-performing AI applications.
- Collaborated with data engineers, software developers, business analysts, and client stakeholders to gather requirements, deliver AI-driven solutions, and support enterprise digital transformation initiatives through data-driven decision-making.

Environment: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers, FastAPI, AWS (S3, EC2, SageMaker, Glue, Lambda), Docker, Kubernetes, Apache Airflow, Tableau, Power BI, Plotly, Git, GitHub.

Education

Master of Business Administration (MBA)

Lewis University, Romeoville, IL- Jan 2024 – May 2025